

Deepak Waman

M.Sc., Ph.D. | Postdoctoral Researcher | Cloud Physics

Institute of Meteorology and Climate Research Troposphere Research (IMKTRO), KIT, Germany

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Research Summary

As an atmospheric scientist, I investigate cloud and precipitation processes and aerosol–cloud–radiation interactions by combining high-resolution numerical modeling with observations from research aircraft, ground-based, and spaceborne instruments. My research focuses on understanding how ice nucleation and multiplication influence cloud glaciation, precipitation efficiency, and cloud radiative properties in convective systems to improve the representation of these processes in atmospheric models. In my postdoctoral research, I integrate, compare, and evaluate cloud microphysical parameterizations within the ICON model by combining observations. I have also been exploring computationally efficient approaches for representing ice multiplication, including machine learning techniques. In parallel, I investigate how aerosols and environmental conditions influence lightning activity across different convective regimes using global observational datasets. My long-term research goal is to understand how cloud microphysical processes and environmental conditions govern thunderstorm electrification and lightning activity, particularly over the Indian monsoon region.

Education

Oct. 2019 – Oct. 2023

Ph.D., Department of Earth and Environmental Sciences, Lund University, Sweden

Thesis: Mechanisms for the Influence from Ice Nucleus Aerosols on Clouds and their Indirect Effects: Cloud Modelling

Supervisor: Dr. Vaughan Phillips

Jul. 2017 – Jun. 2019

M.Sc., Atmospheric Sciences, Savitribai Phule Pune University, Pune, India

74.9%; CGPA 9.0/10

Jun. 2013 – Jun. 2016

B.Sc., Physics, Savitribai Phule Pune University, Pune, India

90.16%

Research and Professional Experience

Dec. 2023 – Present

Postdoctoral Researcher, IMKTRO, Karlsruhe Institute of Technology (KIT), Germany

Supervisor: Prof. Corinna Hoose

Project: WarmWorld (German Federal Ministry for Research, Technology and Space, BMFTR)

Focus: Improving mixed-phase and ice cloud representation in ICON for km-scale climate simulations

Independent research:

- Influence of aerosols, cloud microphysics, and thermodynamic environments on global lightning activity
- Data-driven (machine learning) approaches for representing secondary ice production in climate models

Research Funding and Grants

Sept. 2026

CHARGE: *Characterizing the Aerosol and Microphysical Controls on Thunderstorm Glaciation and Electrification in Present and Future Climate*

DFG Individual Research Grant (Eigene Stelle) – **Principal Investigator** (Under Review)

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8	5	205

First-Author Publications

1. **Waman, D.**, Meusel, J., Keshtgar, B., Wallentin, G., Barthlott, C., Patade, S., Shete, S., Prabhakaran, T., Fievet, R., Finney, D., Blyth, A., & Hoose, C. (2026). Impacts of secondary ice production on the microphysics and dynamics of deep convective clouds in different environments. *Atmospheric Chemistry and Physics*. doi:10.5194/acp-26-11371-2026
2. **Waman, D.**, Jadav, A., Patade, S., Gautam, M., Deshmukh, A., & Phillips, V. (2025). Mechanisms for indirect effects from ice nucleating particles on continental clouds and radiation. *Journal of the Atmospheric Sciences*, **82**(8), 1693–1720. doi:10.1175/JAS-D-24-0143.1
3. **Waman, D.**, Deshmukh, A., Jadav, A., Patade, S., Gautam, M., Phillips, V., Bansemer, A., & Jakobsson, J. (2023). Effects from time dependence of ice nucleus activity for contrasting cloud types. *Journal of the Atmospheric Sciences*, **80**(8), 2013–2039. doi:10.1175/JAS-D-22-0187.1
4. **Waman, D.**, Patade, S., Jadav, A., Deshmukh, A., Gupta, A.K., Phillips, V.T., Bansemer, A., & DeMott, P.J. (2022). Dependencies of four mechanisms of secondary ice production on cloud-top temperature in a continental convective storm. *Journal of the Atmospheric Sciences*, **79**(12), 3375–3404. doi:10.1175/JAS-D-21-0278.1

Co-Author Publications

5. Pant, C.S., **Waman, D.**, Patade, S., Deshmukh, A., Singh, N., Phillips, V. & Bansemer, A. (2026). A modified stratiform cloud microphysics parameterization: evaluation using the Community Atmosphere Model version 6 single-column model. *Atmospheric Chemistry and Physics*, **26**(10), 7407–7433. doi:10.5194/acp-26-7407-2026
6. Jadav, A., **Waman, D.**, Pant, C. S., Patade, S., Gautam, M., Phillips, V., Bansemer, A., Barahona, D., & Storelmov, T. (2025). An improved convection parameterization with detailed aerosol–cloud microphysics for a global model. *Journal of the Atmospheric Sciences*, **82**(1), 197–231. doi:10.1175/JAS-D-23-0175.1
7. Patade, S., Kulkarni, G., Patade, S., **Waman, D.**, Sotiropoulou, G., Samanta, S., Malap, N. & Prabhakaran, T. (2025). Importance of secondary ice production in mixed-phase monsoon clouds over the Indian subcontinent. *Atmospheric Research*, **315**, 107890. doi:10.1016/j.atmosres.2024.107890
8. Gautam, M., **Waman, D.**, Patade, S., Deshmukh, A., Phillips, A., Korczynski, M., Paul, F., Smith, P., & Bansemer, A. (2024). Fragmentation in collisions of snow with graupel/hail: new formulation from field observations. *Journal of the Atmospheric Sciences*, **81**(12), 2149–2164. doi:10.1175/JAS-D-23-0122.1
9. Gupta, A.K., Deshmukh, A., **Waman, D.**, Patade, S., Jadav, A., Phillips, V., Bansemer, A., Martins, J., & Goncalves, F. L. T. (2023). The microphysics of the warm-rain and ice crystal processes of precipitation in simulated continental convective storms. *Communications Earth & Environment*, **4**(1), 226. doi:10.1038/s43247-023-00884-5
10. Patade, S., **Waman, D.**, Deshmukh, A., Gupta, A., Jadav, A., Phillips, V., Bansemer, A., Carlin, J. & Ryzhkov, A. (2022). The influence of multiple groups of biological ice nucleating particles on microphysical properties of mixed-phase clouds observed during MC3E. *Atmospheric Chemistry and Physics*, **22**(18), 12055–12075. doi:10.5194/acp-22-12055-2022
11. Jakobsson, J.K., **Waman, D.**, Phillips, V., & Kristensen, T. (2022). Time dependence of heterogeneous ice nucleation by ambient aerosols: laboratory observations and a formulation for models. *Atmospheric Chemistry and Physics*, **22**(10), 6717–6748. doi:10.5194/acp-22-6717-2022
12. Deshmukh, A., Phillips, V.T., Bansemer, A., Patade, S., & **Waman, D.** (2022). New empirical formulation for the sublimational breakup of graupel and dendritic snow. *Journal of the Atmospheric Sciences*, **79**(1), 317–336. doi:10.1175/JAS-D-20-0275.1

Manuscripts in Preparation

1. **Waman, D.** (2026). Contrasting effects of insoluble aerosols on extreme lightning across continental hotspots. *To be submitted to Geophysical Research Letters*.

2. **Waman, D.**, Nassar, A., & Hoose, C. (2026). Aerosols and ice fraction govern global lightning. *To be submitted to Atmospheric Chemistry and Physics*.

Invited Talks

4 Sept. 2026

“Multiscale Effects of Extra Ice in Clouds,” 6th Early Career Research Seminar, Indian Institute of Tropical Meteorology (IITM), Pune, India.

youtube.com/live/J7TY2_K6GcM

Selected Conferences and Workshops

May 2026

European Geoscience Union (EGU), Vienna, Austria

Poster: Regime-dependent impacts of CCN and cloud glaciation on global lightning activity. [doi:10.5194/egusphere-egu26-3745](https://doi.org/10.5194/egusphere-egu26-3745)

Apr. 2026

Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria

Apr. 2024

European Geoscience Union (EGU), Vienna, Austria

Talk: Importance of age of convective clouds for explosive ice crystal number growth via secondary ice production. [doi:10.5194/egusphere-egu24-2859](https://doi.org/10.5194/egusphere-egu24-2859)

Poster: Sensitivity of microphysical properties of mixed-phase clouds on model resolution and microphysics scheme in ICON. [doi:10.5194/egusphere-egu24-11022](https://doi.org/10.5194/egusphere-egu24-11022)

Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria

Jun. 2024

ICON Numerical Model Training Course, Deutscher Wetterdienst (DWD), Offenbach, Germany

Apr. 2023

European Geoscience Union (EGU), Vienna, Austria

Poster: Singular versus time-dependent approach of heterogeneous ice nucleation and secondary ice production in contrasting clouds. [doi:10.5194/egusphere-egu23-1834](https://doi.org/10.5194/egusphere-egu23-1834)

Dec. 2022

American Geophysical Union (AGU), Chicago, USA

Oral: Primary and secondary ice formation in various clouds: a numerical study

Aug. 2022

American Meteorological Society Collective Madison Meeting, Wisconsin, USA

Poster: Role of four SIP mechanisms in ice enhancement in continental convective clouds: a numerical study

Aug. 2021

International Commission for Clouds and Precipitation, IITM, Pune, India

Poster: Dependency of ice enhancement ratio on cloud-top temperature in continental deep convective clouds

Teaching Experience

Oct. 2025 – Feb. 2026

Lecturer and Tutor, Cloud Physics (M.Sc. level), KIT, Germany

Designed and delivered lectures on Parcel Theory, Cloud Microphysics, and Storm Dynamics to 20 M.Sc. students. Developed Mentimeter-based interactive quizzes for in-class assessment.

Oct. 2024 – Feb. 2025

Tutor, Cloud Physics (M.Sc. level), KIT, Germany

Conducted problem-solving sessions for 14 M.Sc. students.

Teaching Interests

- Cloud Physics
- Atmospheric Thermodynamics and Dynamics
- Numerical and Computational Techniques in Atmospheric Science
- Atmospheric Electricity and Thunderstorm Electrification
- Computational laboratory (Fortran, Python) and Numerical Modeling (ICON, WRF)

Student Supervision

2026-Ongoing

Ann Henze – M.Sc. thesis: Data-driven approach to understand controls on lightning occurrence across global lightning hotspots (Main Supervisor, KIT)

2026-Ongoing

Xuan Zhang – M.Sc. thesis: Predicting ice multiplication using machine learning (Main Supervisor, KIT)

2026-Ongoing

Antonis Michalopoulos – M.Sc. thesis: Impact of secondary ice production on the dynamics of extratropical cyclones (Co-supervisor, KIT)

2025-Ongoing

Abdullah Nassar – Research assistant: Influence of soluble aerosols on lightning activity over Europe (Main Supervisor, KIT)

Professional Service

Peer reviewer for:

- *Atmospheric Chemistry and Physics* (EGU/Copernicus)
- *Journal of Geophysical Research – Atmospheres* (AGU)
- Springer Nature journals
- *Advances in Atmospheric Sciences* (Science Press and Springer)

Referees

Prof. Corinna Hoose (Professor, Postdoc Advisor)

Institute of Meteorology and Climate Research Troposphere Research (IMKTRO),
Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany
corinna.hoose@kit.edu

Dr. Vaughan Phillips (Senior Lecturer, Ph.D. Advisor)

Department of Earth and Environmental Sciences
(Formerly Dept. of Physical Geography and Ecosystem Science),
Lund University, Sweden
vaughan.phillips@mgeo.lu.se

Prof. Chandra Shekhar Pant (Assistant Professor)

Department of Hydro & Renewable Energy
Indian Institute of Technology (IIT), Roorkee, India
csp@hre.iitr.ac.in